MIL-C-63413A(AR)
24 October, 1983
SUPERSEDING
MIL-C-63413(AR)
8 September 1981

MILITARY SPECIFICATION

CARTRIDGE, 20MM, HIGH EXPLOSIVE, INCENDIARY, TRACER, SELF DESTRUCT, M246A1 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, preparation for delivery for one type of cartridge designated as Cartridge, 20MM, High Explosive, Incendiary, Tracer, Self Destruct, M246A1 for use in Cannon, 20MM, Automatic, M168.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications For

STANDARD

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC: 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-644	-	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking for Small Arms Ammunition
MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (SEE 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

7258801	-	Classification of Defects for Case, Cartridge, 20MM, M103, M103A1, M103B1.
9333764	-	Cartridge, 20MM, High Explosive, Incendiary, Tracer, Self Destruct, M246A1.
L19333764	-	Index of Inspection Equipment Lists for: Cartridge, 20mm, High Explosive, Incendiary Tracer-Self Destruct, M246A1.
9333765	-	Projectile, 20mm, HEIT-SD for the M246A1 Cartridge.
9333766	-	Charged Body, Projectile, 20mm, HEIT-SD for the M246A1 Cartridge.
10542428	-	Disc, Metering.
10542429	-	Seal.

Inspection Equipment Drawings

7259424	-	Gage, Torque Testing.
7259443	-	Rack, Cartridge.
7296962	-	Indicator Gage.
7555162	-	Gage, Chamber.
7556901	-	Indicator, Comparator Gage.
8655181	-	Alignment Check Taper Plug Gage.
8655592	-	Ring Gage, GO, Special.
11075013	-	Gage, Feeler.
11075060	-	Gage, Indicator.
11075126	-	Test Fixture (Projectile Extraction).
11075284	-	Primer Resistant Test Circuit.

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PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

a Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.13).

3.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg.) 9333764, all associated drawings, and with all requirements specified in applicable specifications.

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a. Projectile pull test. The force necessary to extract the projectile from the cartridge case shall be 1100 and 2800 pounds, inclusive.

b. Projectile torque. The projectile shall withstand a minimum torque of 10 inch pounds without movement.

3.3 Contamination of energetic components. Extreme care shall be exercised to avoid contamination of primers and propellant by oil, grease or other foreign matter since such contamination can cause misfires or increase action time and lot rejection.

3.4 Electrical resistance of primed case. The electrical resistance of the primed case shall not be less than 500 ohms or more than 1,200,000 ohms.

3.5 Functioning. The cartridge shall comply with the following:

3.5.1 Conditioned at 68°F to 72°F.

a. Velocity. Average velocity of the projectile shall be 3380 ± 50 feet per second (fps) at 78 feet from the muzzle of the gun and a standard deviation shall not exceed 40 fps. (See 6.6.)

b. Pressure. Average chamber pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) (see 6.7) of the chamber pressure shall not exceed 60,500 pounds per square inch (psi).

c. Waterproofness. The average velocity (wet) of projectiles shall not vary from the average velocity (dry) of projectiles by more than 100 fps.

3.5.2 Conditioned at -65°F.

a. Action time: The action time of the sample cartridges and the average action time (X) plus four sample standard deviation (S) shall not exceed 4.0 milliseconds (see 6.6).

b. Projectile trace: The projectile shall exhibit a trace visible against a dark background for a minimum of 3.1 seconds unless self-destruction occurs earlier.

c. Projectile Self-Destruct: The projectile shall self-destruct between 3.1 and 7 seconds (inclusive) of un-interrupted flight.

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3.5.3 Conditioned at ambient temperature.

a. Function and casualty. Cartridges shall function without casualty when fired in the automatic weapons.

b. Projectile plate function. Projectile shall function with high order detonation upon target impact. (Chapter 1, Section 6, AMCR 715-505, Vol 8 and See 6.8 and 6.10).

c. Accuracy. Average mean radius of all targets shall not exceed 15 inches (accuracy) at a range of 600 yards.

3.5.4 Conditoned at +145°F.

a. Projectile trace: The projectile shall exhibit a trace visible against a dark background for a minimum of 3.1 seconds unless self-destruction occurs earlier.

b. Projectile Self-Destruct. The projectile shall self-destruct between 3.1 and 7 seconds (inclusive) of un-interrupted flight.

3.5.5 Safety. There shall be no premature burst (see 6.11) or evidence thereof nor any projectile sticking in the gun bore in any of the test firings.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.7.1 Processing defects. The metal parts and assemblies shall be free of cracks, splits, ruptures, perforations, dents, deformities, exposed explosive and other forms of substandard workmanship.

3.7.2 Cleanliness. All components and assemblies shall be free of foreign matter, such as oil, grease, corrosion, dirt, chips, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (4.3)
b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Seal	10542429	125
Metering Disc	10542428	125
Charged Body, Projectile	9333766	125
Projectile Assembly	9333765	125
Cartridge	9333764	700
Packaging	(See 5.1)	7

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein will result in sample disapproval.

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TABLE I. FIRST ARTICLE INSPECTION

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Cartridge and Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Seal (Dwg. 10542429) Examination for defects.	125		3.2	4.4.2.1
	Metering Disc (Dwg. 10542428) Examination for defects.	125		3.2	4.4.2.2
	Charged Body, Projectile (Dwg. 9333766) Examination for defects.	125		3.2	4.4.2.3
	Projectile Assembly (Dwg. 9333765) Examinaton for defects.	125		3.2	4.4.2.4
	Cartridge (Dwg. 9333764) Examination for defects Test requirements.	700 700		3.2 3.2	4.4.2.5 4.4.2.6
	Packaging Examination for defects.	7		3.1.a	4.4.4/5.1

NOTES:

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectile parts from one interfix lot number from one manufacturer.

c. Fuze assemblies from one interfix lot number from one manufacturer.

d. Primers from one interfix lot number from one manufacturer.

e. Propellant from one lot number from one manufacturer.

f. Incendiary pellets from one interfix lot number from one manufacturer.

g. High explosive from one interfix lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.50 percent and the AQL for minor defects shall be 2.50 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, where three or less defects appear in a classification sampling plans applicable to a class shall not be used for major defects.

c. Primed Cartridge Cases. Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.5.4.1. Primed cases which fail to meet the requirement of 3.4 shall be removed from the lot.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10542429
4.4.2.1	Seal				NEXT HIGHER ASSEMBLY 9333766
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		.25%	3.2	Gage
102	Corrosion or foreign matter		.25%	3.2	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10542428
4.4.2.2	Metering Disc				NEXT HIGHER ASSEMBLY 9333766
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	See Dwg. 10542428				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9333766
4.4.2.3	Charged Body, Projectile				NEXT HIGHER ASSEMBLY 9333765
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Depth from metering disc seat to tracer igniter, max (prior to insertion of seal and metering disc)		100%	3.2	Gage
2	Explosive on threads at indicated length		100%	3.2	Visual
3	Depth, mouth to HE mix		100%	3.2	Gage
4	Seal missing or perforated		100%	3.2	Visual
Major					
101	Self-Destruct propellant missing (prior to pellet insertion)		100%	3.2	Visual
102	Incendiary pellet missing (prior to high explosive charging)		100%	3.2	Visual
103	Depth from metering disc seat to tracer igniter, min (prior to insertion of seal and metering disc)		.25%	3.2	Gage
104	Metering disc missing		.25%	3.2	Visual
105	Metering disc crimp defective		.25%	3.2	Visual/Manual 1/
106	Bourrelet diameter, max		.25%	3.2	8655592
107	Foreign matter		.25%	3.7	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

See footnotes following 4.4.2.6.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63413A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile Assembly	1 OF 1	9333765
			NEXT HIGHER ASSEMBLY 9333764

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Disassembly torque test of fuze		.25%	3.2	Gage/4.5.1 2/ 3/
Minor					
201	Gap, max, between fuze and body		.40%	3.2	11075013 2/
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:
See footnotes following 4.4.2.6

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.5	Cartridge, 20mm, HEIT-SD, M246A1 4/ (Dimensional Requirements)	1	2	9333764
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Split or perforated case		100%	3.2	Visual
2	Crack or split in projectile		100%	3.2	Visual
3	Case to projectile crimp missing		100%	3.2	Visual
4	Low propellant charge		100%	3.2	Gage 5/
Major					
101	Chamber gage failure (profile & alignment, max)		100%	3.2	7555162/8655181 6/
102	Depth of primer seating		.25%	3.2	7296962
103	Total length, max		.25%	3.2	7556901
104	Length from case head to centerline of case crimp, max		.25%	3.2	11075060
105	Fuze nose cap damaged		.25%	3.2	Visual
106	Loose primer		.25%	3.2	Manual
107	Corrosion		.25%	3.2	Visual
108	Head configuration		.25%	3.2	Visual
109	Metal sliver(s) on head face		.25%	3.2	Visual
110	Improper or incomplete crimp (case-projectile)		.25%	3.2	Visual

NOTES:

See footnotes following 4.4.2.6.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cartridge, 20mm, HEIT-SD, M246A1 (Dimensional requirements) 4/	2 OF 2	9333764
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Waterproofing lacquer missing around primer		.40%	3.2	Visual
202	Lacquer on primer button		.40%	3.2	Visual
203	Primer crimp missing or incomplete		.40%	3.2	Visual
204	Rotating band damaged		.40%	3.2	Visual
205	Cartridge case damaged		.40%	3.2	Visual
206	Marking misleading or unidentifiable		.40%	3.2	Visual
207	Protective coating damaged or incomplete		.40%	3.2	Visual
208	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:
See footnotes following 4.4.2.6

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cartridge, 20mm, HEIT-SD, M246A1 (Test requirements) 4/	1 OF 2	9333764
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Test Requirements				
	Functioning				
	(1) Conditioned at 68°F to 72°F			3.5.1	4.4.3.1/4.4.3.6/4.5.5
	a. Velocity	20		3.5.1a	4.4.3.6a/4.5.5.1 7/
	b. Pressure	20		3.5.1b	4.4.3.6b/4.5.5.1 7/
	c. Waterproofness	20		3.5.1c	4.4.3.6c/4.5.5.1
	(2) Conditioned at -65°F			3.5.2	4.4.3.1/4.5.6.1
	a. Action time	50		3.5.2a	4.5.6.2
	b. Projectile Trace	110		3.5.2b	4.4.3.4/4.5.6.3 8/
	c. Projectile Self-Destruct	110		3.5.2c	4.4.3.4/4.5.6.3 8/
	(3) Conditioned at ambient temperature			3.5.3	4.4.3.1/4.5.7
	a. Function and casualty	200		3.5.3a	4.4.3.2/4.5.7.2
	b. Projectile plate function	50		3.5.3b	4.4.3.3/4.5.7.1
	c. Accuracy	50		3.5.3c	4.4.3.7/4.5.7.3
	(4) Conditioned at +145°F			3.5.4	4.4.3.1/4.5.8.1
	a. Projectile Trace	110		3.5.4a	4.4.3.4/4.5.8.2 8/
	b. Projectile Self-Destruct	110		3.5.4b	4.4.3.4/4.5.8.2 8/

NOTES:

See footnotes following 4.4.2.6

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cartridge, 20mm, HEIT-SD, M246A1 (Test requirements) 4/	2 OF 2	9333764
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Static				
	Pull test of projectile	20		3.2a	4.4.3.5/4.5.3 9/ 11075126/7259443
	Torque test of projectile	20		3.2b	4.4.3.8/4.5.2/7259424
	Propellant contamination/charge weight verification	20	0-1	3.3	4.5.3.1 9/
	Electrical resistance of primer verification)	20	0-1	3.4	4.5.4.2 9/

NOTES:

See footnotes following 4.4.2.6

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Footnotes

1/ If the crimp closure is not pressed tightly against the metering disc for 360° or the metering disc can be moved manually, the charged body sub-assembly shall be classed defective.

2/ May be performed on cartridge.

3/ When fuze turns with respect to the projectile body upon application of the specified torque in a counter clockwise direction, the cartridge shall be classed defective.

4/ MIL-STD-651 and Dwg. 7258801 shall apply in defining and evaluating cartridge visual defects.

5/ A cartridge containing 550 grains or less of propellant shall be classed defective.

6/ Army design 7555162 shall be utilized for gaging cartridge using M103 (brass) case only. Cartridges utilizing the M103A1 (steel) case shall be checked with a gage designed by the contractor. A dead weight load of 60 pounds may be used to insert the cartridge in the gage.

7/ Velocity and pressure readings shall be recorded in a single test.

8/ In each temperature phase, projectile trace and projectile self-destruct are to be conducted simultaneously.

9/ Tests may be conducted using same sample cartridges.

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4.4.3 Tests. The tests listed in 4.4.2.6 shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridges which have met the visual and dimensional requirements and which have been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be mixed thoroughly before being divided into samples for the various tests.

4.4.3.1 All cartridge firing tests. The lot shall be rejected if in any firing test one or more of the following casualties occur:

a. Misfire associated with e or f of paragraph 6.12.

b. Projectile remaining in bore.

c. Metal parts seperation, except rotating band separation. (See 6.9.2)

d. Premature projectile functioning (see 6.11).

4.4.3.2 Function and casualty. Two-hundred cartridges shall be subjected to this test. The lot shall be rejected if, at any time, any of the following occurs:

a. Any fire "out of battery" (See 6.9.1).

b. Any primer misfires. (See 6.12)

c. Any cartridge case fails to extract.

d. Any cartridge case longitudinal split occurs (L or M) (Dwg. 7258801).

e. Any cartridge case circumferential rupture, partial (L) or complete (Dwg. 7258801).

f. Any detached metal from cartridge case occurs (See 6.9.3).

g. Any cartridge fails to chamber.

In addition, if the following defects occur equal to or less than the "ACC" number, the lot shall be accepted. If any defect(s) occur equal to or exceeding the "REJ" number, the lot shall be rejected. If any defect(s) occur in excess of the "ACC" number, but less than the "REJ" number, a second sample of 400 cartridges shall be selected and fired in the M168 automatic cannon. The lot shall be rejected if any defect occurs equal to or greater than the "CUMULATIVE REJ" number.

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		ACC	REJ	CUMULATIVE REJ
1.	Primer missing or button missing from primer	0	2	2
2.	Primer leak (see NOTE 1)	9	20	25
3.	Longitudinal case split (I or S) (Dwg. 7258801)	10	16	31
4.	Longitudinal case split (K) (Dwg. 7258801)	0	2	2
5.	Case circumferential rupture, partial (S, J, or K) (Dwg. 7258801)	4	9	11
6.	Longitudinal case split (J) (Dwg. 7258801)	4	9	11
7.	Projectile rotating band separation complete or partial separation having band seat serration imprint. (See 6.9.4)	0	3	3
8.	Projectile rotating band separation having greater than 0.5 inch and without band seat serration imprint. (See 6.9.4)	1	4	6

NOTE 1. For primer defect definitions, see Chapter 1, Section 3, Paragraph 3.6 of AMCR 715-505, Volume 8.

4.4.3.3 Projectile plate function. Fifty cartridges shall be subjected to this test. The lot shall be rejected if five or more projectiles fail to detonate high order upon impact. The lot shall be accepted if not more than two projectiles fail. If more than two or less than five projectiles fail, an additional fifty cartridges shall be tested for the projectile plate function parameter only. If the cumulative number of failures equals or exceeds 7, the lot will be rejected.

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4.4.3.4 Projectile trace and self-destruct. For acceptance each temperature phase shall be considered individually. If a total of 10 or more sample cartridges have projectiles that fail to comply with trace or self-destruct requirements, the lot shall be rejected. If more than six and less than 10 cartridges fail, a second sample of 110 shall be tested. The lot shall be rejected if sixteen (16) or more cartridges of the combined samples fail to meet the requirements.

4.4.3.5 Pull test of projectile.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined sample fail to comply with the maximum requirement.

4.4.3.6 Cartridge functioning at 68°F to 72°F. If any requirement(s) in paragraph 3.5.1 is not achieved on the first sample, a second sample will be tested for the failing parameter(s).

a. Velocity. If the sample fails to comply with either or both requirements a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the retest sample fails to comply with the applicable requirement(s). (See 6.6)

b. Pressure. If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 60,500 psi, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirement. (See 6.7)

c. Waterproofness. If the average velocity (wet) of the sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirements of 3.5.1.a.

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4.4.3.7 Accuracy. If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of 10 retest accuracy targets exceeds the applicable requirements (see 3.5.3.c).

4.4.3.8 Projectile Torque. Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

4.4.4 Packaging, packing and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644.

4.4.5 Inspection equipment. The inspection equipment required to perform the tests and examination prescribed herein is described in the Index of Inspection Equipment List LI9333764 and the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Disassembly torque test of fuze on projectile. After six (6) hours curing time, the projectile shall be placed in the applicable fixture and the inch pound requirement specified on the applicable drawing shall be applied. Observation shall be made for the requirement specified on the applicable drawing. This test is a non-destructive test.

4.5.2 Torque test of projectile. The cartridge shall be placed in the applicable fixture specified herein and the inch pound torque specified on the applicable drawing shall be applied slowly to the projectile. The cartridge shall be marked with a light scratch extending axially across the rotating band onto the case neck. Observation shall be made for movement of the projectile with respect to the misalignment in the scratch mark. This test is a non-destructive test.

4.5.3 Pull test of projectile. The cartridge shall be placed in the applicable fixture specified herein and the tension force specified on the applicable drawing shall be applied. Projectile must separate from the cartridge case in the range specified on the applicable drawing. The method of test shall be as specified in Chapter 1, Section 10 of AMCR 715-505, Volume 8. This test is a destructive test. Primed cartridge cases may be used for the electrical resistance test specified in 4.5.4.2.

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4.5.3.1 Propellant contamination and charge weight verification. After completion of test specified in 4.5.3, each of the cartridges tested will be examined visually for contamination as described in Chapter 1, Section 10, AMCR 715-505, Volume 8. The charge from each of the sample cartridges shall be weighed.

4.5.4 Electrical resistance.

4.5.4.1 Electrical resistance of primed case. The primed case shall be tested for the applicable requirement using Government approved equipment and circuitry specified on Dwg. 11075284 to measure the resistance of the primer.

4.5.4.2 Verification. A Simpson Volt - Ohm - Ammeter, Model 260, or approved equal, will be used to measure the resistance of the primer in each primed cartridge case from the projectile pull test. Only the RX 100 and RX 10,000 scales of the meter will be used in order to limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milliamperes, maximum.

4.5.5 Functioning (conditioned at 68°F to 72°F).

4.5.5.1 Velocity, pressure and waterproofness. These tests shall be conducted in accordance with AMCR 715-505, Volume 8, Chapter 1, Section 2. Standard deviation and range shall be computed in accordance with 6.6 and 6.7 respectively.

4.5.6 Functioning (conditioned at -65°F).

4.5.6.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at -65°F ± 5°F. Timing for the conditioning period shall start when the chamber has stabilized following the sample being placed in the conditioning chamber.

4.5.6.2 Action time. The method of test shall be as prescribed in AMCR 715-505, Volume 8.

4.5.6.3 Projectile trace and self-destruct. The method of test shall be as specified in AMCR 715-505, Volume 8, Chapter 1, Section 8. Test for projectile trace shall be conducted simultaneously with self-destruct. Projectile trace and self-destruct time will begin at cartridge ignition and continue until end of the tracer burn or at self-destruct whichever comes first. Observation shall be from near the gun position.

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4.5.7 Functioning (conditioned at ambient temperature).

4.5.7.1 Projectile plate function. This test shall be conducted in accordance with AMCR 715-505, Chapter 1, Section 6, Volume 8. The aluminum target (2024 T4, T3 or T36) of .040 inch nominal thickness shall be located 200 ± 25 yards from the muzzle of the gun normal to the line of fire. Observation of projectile functioning shall be made from the camera position.

4.5.7.2 Function and casualty. The method of testing shall be in accordance with AMCR 715-505, Chapter 1, Section 3, Volume 8. The test sample shall be fired in bursts of 50 cartridges in the M168 cannon. The weapon barrel(s) shall be at ambient temperature at the beginning of test and cooled to ambient after each 100 rounds.

4.5.7.3 Accuracy. The method of test and measurement of targets shall be in accordance with AMCR 715-505, Volume 8, Chapter 1, Section 4. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards, the average mean radius shall be reduced in direct proportion to the reduction in range. The 200 yard range may be reduced by 11.5 yards to accommodate combining the accuracy and projectile plate functioning tests. Any variation from the 200 yard range requires direct proportional adjustment of the average mean radius (see 6.10).

4.5.8 Functioning (conditioned at +145°F).

4.5.8.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at +145°F ± 5°F. Timing for the conditioning period shall start when the chamber has stabilized following the sample being placed in the conditioning chamber.

4.5.8.2 Projectile trace and self-destruct. (See 4.5.6.3.)

4.5.9 Defective weapon. If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or is found to be defective during or at the end of burst, the test shall be suspended and Government approval sought for a new test. The cyclic rate of each burst shall not be less than 3,000 shots per minute in the gun. The cyclic rate shall be recorded.

4.5.10 Test validity. If for any reason the test activity considers that the test conditions have detrimentally affected the test results, the test activity may request the Government to declare the test invalid and authorize a new test.

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5. PACKAGING

5.1 Preservation and packaging. The M246A1 cartridge shall be preserved and packaged in accordance with the instructions and the applicable Combination of Adopted Items Drawings from the procuring agency.

5.2 Packing. The cartridge shall be packed in accordance with the Combination of Adopted Items Drawing from the procuring agency.

5.3 Marking. Marking of the packed container shall be in accordance with the applicable Combination of Adopted Items Drawing and Code of Federal Regulations, Title 49, Parts 100-199.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. This specification covers the loading, assembling and packing for Cartridge, 20MM, High Explosive, Incendiary, Tracer, Self-Destruct, M246A1 for use in the M168 automatic cannon systems which are designed to utilize electric primed cartridges with M103 type case.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRSMC-QAF-I(D), U.S. Army Armament Munitions and Chemical Command (AMCCOM), Dover, NJ, 07801.

6.4 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by AMCCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, AMCCOM, ATTN: DRSMC-QAF-S(D) and DRSMC-SCA-CE(D), Dover, NJ, 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of AMCCOM.

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6.6 Standard deviation. Standard deviation shall be calculated from the following formula or other approved formula:

$$S = \sqrt{\frac{\Sigma(Xi - \bar{X})^2}{n}}$$

where:

Xi = Each individual value

$\bar{X}$ = Sample arithmetic mean = $\frac{\Sigma Xi}{n}$

n = Sample size

$\Sigma(Xi - \bar{X})^2$ = the sum of the squares of the differences between the sample mean and each individual value

6.7 Range plan. Where a range plan is specified, the individual values will be recorded in subgroups of 5 in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range ($\bar{R}$).

6.8 Projectile function. The photographic standards shown in Chapter 1, Section 6, Para 6.6 of AMCR, 715-505, Volume 8 with supplement are applicable for comparison of the projectile performance for the HEI cartridge functioning test. Photographic standards should be requested from the technical activity for 20MM, HEI, Cartridges.

6.9 Defect definitions.

6.9.1 "Out of battery". A cartridge shall be classed defective if it fires, due to delayed ignition, after the bolt has been unlocked and extraction has been initiated or completed.

6.9.2 Metal part separation. Separation or breakup of projectile part(s), except rotating band separation, as evidenced by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s).

6.9.3 Detached metal from cartridge case. Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears. Shavings of metal from the interior wall of the case neck, in the crimped area, shall not be considered as detached metal.

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6.9.4 Rotating band separation. For classification as a defect, there must be evidence by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). Normal band fringing shall not be classed as a defect.

6.10 Testing precaution. If an open projectile impact area is in use at the test facility, precaution should be used because of the fire hazard.

6.11 Premature burst. A premature burst is considered to have occurred if the round functions in the gun bore or during uninterrupted flight at a distance less than 200 yards from the gun muzzle.

6.12 Misfires. Misfiring test cartridges, excepting those from the waterproofness test shall be retested in equipment identified in accuracy test. If a cartridge fails to function in that test, examinations shall be conducted for the following conditions:

a. Excessive primer seating depth.

b. Metal slivers across primer insulator.

c. Electrical continuity.

d. Missing or obstructed case vent.

e. Missing or insufficient primer composition or propellant.

f. Partially burned or contaminated primer composition or propellant.

The lot shall not be penalized if condition a. is observed. However, this condition shall be reported for information. The "Acc" number shall be zero (0) and the "Rej" number 3, for misfires associated with conditions b, c, or d. The second sample for evaluation of misfires associated with conditions b, c, or d shall consist of the function and casualty second sample (see 4.4.3.2). The lot shall be rejected if either condition e or f is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volumn 8.

6.13 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

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6.14 Safety data. For safety data requirements refer to Safety Data Document as listed on Product Drawings.

6.15 First article sample. In order to minimize inspection costs, the firing tests of 4.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the test facility.

6.16 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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